

Goal: run through the distribution theory behind the ANOVA.

The sum of square error is

$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

where n_i is the size of the i th group and $\bar{X}_i$ the mean of the i th group.

Theorem If (1) X_{ij} is an independently and normally distributed random variable with mean μ_i and variance σ^2 , and (2) W_i^2 is the sample variance of the observations in the i th group where

$$W_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2,$$

then

$$\frac{SSE}{\sigma^2} \sim \chi_{n-k}^2$$

Proof Recall from Stat 115A that, if those conditions hold, then

$$\frac{(n_i - 1)S_i^2}{\sigma^2} \sim \chi_{n_i-1}^2$$

Further, the sum of multiple chi square random variables results in another chi square random variable, with degrees of freedom equal to the sum of the degrees of freedom from the added random variables. So consider

$$\sum_{i=1}^k \frac{(n_i - 1)W_i^2}{\sigma^2} = \frac{\sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2}{\sigma^2} = \frac{SSE}{\sigma^2}$$

and since the observations are independent, this is the sum of multiple chi-square variables with degrees of freedom $n_i - 1$. So this follows a chi-square distribution with $\sum_{i=1}^k (n_i - 1) = n - k$ degrees of freedom. ■

The mean square error is

$$MSE = \frac{SSE}{df_E} = \frac{SSE}{n - k}$$

Theorem The mean square error is an unbiased estimator of σ^2 .

Proof We want to show that $E(MSE) = \sigma^2$. And we know that the expected value of a chi-square distribution is its degrees of freedom. So

$$E\left[\frac{SSE}{\sigma^2}\right] = n - k$$

and

$$E[MSE] = E\left[\frac{SSE}{n - k}\right] = E\left[\frac{\sigma^2}{n - k} \frac{SSE}{\sigma^2}\right] = \frac{\sigma^2}{n - k} (n - k) = \sigma^2$$

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The sum of squares group (or treatments) is

$$SSG = \sum_{i=1}^k (\bar{X}_i - \bar{X})^2$$

where k is the number of groups. This quantifies the error remaining after explaining some of the variation by the treatment means. Further, the mean square group

$$MSG = \frac{SSG}{df_G} = \frac{SSG}{k - 1}$$

Theorem If all the μ_i are equal, then

$$\frac{SSG}{\sigma^2} \sim \chi_{k-1}^2$$

Theorem The mean square group is an unbiased estimator of σ^2 only if the null hypothesis is true (all k group means are equal).

Proof Consider $E(SSG)$. First, we can rewrite SSG as

$$SSG = \sum_{i=1}^k n_i \bar{X}_i^2 - n \bar{X}^2$$

(algebra supplied by the enthusiastic student) so then

$$E(SSG) = E \left[\sum_{i=1}^k n_i \bar{X}_i^2 - n \bar{X}^2 \right] = \left[\sum_{i=1}^k n_i E(\bar{X}_i^2) \right] - n E(\bar{X}^2)$$

Then, because we can write $E(X^2) = Var(X) + \mu^2$, we can write

$$E(SSG) = \left[\sum_{i=1}^k n_i \left(\frac{\sigma^2}{n_i} + \mu_i^2 \right) \right] - n \left[\frac{\sigma^2}{n} + \bar{\mu}^2 \right]$$

where $\bar{\mu} = \frac{1}{n} \sum_{i=1}^k n_i \mu_i$, the weighted mean of the group means, because

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} E(X_{ij}) = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} \mu_i = \frac{1}{n} \sum_{i=1}^k n_i \mu_i = \bar{\mu}$$

Finally, simplifying we get

$$E(SSG) = \left[\sum_{i=1}^k \sigma^2 \right] + \left[\sum_{i=1}^k n_i \mu_i^2 \right] - \sigma^2 - n \bar{\mu}^2 = \sigma^2(k-1) + \left[\sum_{i=1}^k n_i (\mu_i - \bar{\mu})^2 \right]$$

If the null hypothesis is true, then $\mu_1 = \mu_2 = \dots = \mu_k = \bar{\mu}$ and

$$E(SSG) = \sigma^2(k-1) + 0$$

so then

$$E(MSG) = E \left[\frac{SSG}{k-1} \right] = \frac{\sigma^2(k-1)}{k-1} = \sigma^2$$

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Note: If the null hypothesis is not true, then we can use this same argument to show that $E(MSG) > \sigma^2$.

Consider

$$F = \frac{MSG}{MSE}$$

Based on what we've shown, (1) if the null is true, then this ratio should be close to 1 and (2) if the alternative is true, the ratio should be greater than 1.

Recall from Stat 115A, that the ratio of two independent chi-square distributions, each divided by its degrees of freedom, follows an F distribution, so

$$F = \frac{MSG}{MSE} = \frac{SSG/(k-1)}{SSE/(n-k)} = \frac{\frac{SSG}{\sigma^2}/(k-1)}{\frac{SSE}{\sigma^2}/(n-k)} \sim F(k-1, n-k)$$

and we will reject the null hypothesis for large values of F. That is, if $F \geq F_\alpha(k-1, n-k)$ or, equivalently, using p-values $P(F(k-1, n-k) \geq F)$